

Original Article

Outcome evaluation of pharmacist-physician collaborative protocol-based antimicrobial treatment for hospitalized stroke patients with aspiration pneumonia

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ARTICLE INFO

Keywords:

Antimicrobial stewardship
Aspiration pneumonia
Inpatient mortality
Length of hospital stay
Pharmacist-physician collaborative protocol-based antimicrobial treatment

ABSTRACT

Objectives: To evaluate the efficacy and patient outcomes of pharmacist-physician collaborative protocol-based antimicrobial treatment regimens for antimicrobial stewardship.

Methods: Patients treated for aspiration pneumonia due to stroke within 48 h after admission to Kochi Medical School Hospital (January 2019 to December 2022) were included. Primary outcomes were the cumulative number of days of antimicrobial treatment and length of hospital stay. Secondary outcomes included the percentage of patients under-dosed with first-choice antimicrobial agents and inpatient mortality.

Results: Group A (66 patients) did not receive the antimicrobial treatment protocol, whereas group B (46 patients) did. There were no differences in the patient backgrounds. Group B had a significantly lower percentage of patients who were undertreated with the first-choice antimicrobial agent (9.1 % vs. 42.9 %). There was no significant difference in inpatient mortality between group A and group B (6.1 % vs. 4.3 %). The cumulative number of days of antimicrobial administration and the length of hospital stay were significantly lower in group B: 7.0 days (95 % CI, 6.0–8.0) vs. 9.0 days (95 % CI, 8.0–11.0) for antimicrobial administration, and 28.5 days (95 % CI, 22.0–35.0) vs. 43.0 days (95 % CI, 28.0–55.0) for hospital stay.

Conclusions: Protocol-based antimicrobial treatment for aspiration pneumonia supports appropriate antimicrobial usage and improves patient quality of life. These findings will assist in the effective treatment of aspiration pneumonia in an aging society.

1. Introduction

Pneumonia in patients undergoing treatment in hospitals in Japan is typically classified as hospital-acquired pneumonia (HAP), ventilator-associated pneumonia, or nursing and healthcare-associated pneumonia [1–3]. Older adults are the most affected group, with aspiration pneumonia being the most common [4]. HAP caused by aspiration has a higher isolation frequency of resistant bacteria than community-acquired pneumonia [5,6]; thus, patients with HAP should

be administered an antimicrobial drug regimen that considers the risk of resistant bacteria. However, the causative organisms may be unknown in many cases during the initial treatment of aspiration pneumonia [7, 8], leaving room for debate regarding the choice of antimicrobial agents. The frequency of aspiration pneumonia is particularly high in stroke patients [9]. This condition significantly impacts functional prognosis and mortality, highlighting the importance of managing it from the early stages of treatment [10,11]. Furthermore, proper selection and administration of antimicrobial agents in the early stages of aspiration

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<https://doi.org/10.1016/j.jiac.2024.09.017>

Received 31 July 2024; Received in revised form 17 September 2024; Accepted 19 September 2024

Available online 21 September 2024

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pneumonia treatment is crucial. In this study, we evaluated the impact of an antimicrobial treatment protocol on patient outcomes. This protocol, developed by a team of pharmacists, infectious disease specialists, and neurosurgeons, aimed to assist in practicing effective antimicrobial treatment and improving patient outcomes.

2. Material and methods

2.1. Antimicrobial treatment protocol for aspiration pneumonia

The Antimicrobial Stewardship Team (AST) was composed of pharmacists with expertise in drug treatment and infectious disease specialists. A treatment protocol was collaboratively developed through consensus among these professionals and neurosurgeons. We investigated the isolation rates of *Pseudomonas aeruginosa* and methicillin-resistant *Staphylococcus aureus* (MRSA) from respiratory specimens (sputum, endotracheal sputum, bronchial lavage fluid, pharyngeal mucus, nasal cavity, oral cavity, and pleural fluid) of patients with stroke admitted to our hospital between January 2018 and December 2021. Our findings were compared with the *P. aeruginosa* isolation rates from respiratory specimens in the Japan Nosocomial Infections Surveillance (JANIS) conducted by the Ministry of Health from January 2018 to December 2021. The isolation rates were calculated as follows: isolation rate (%) = (number of isolates from target respiratory specimen/total number of respiratory specimen isolates) × 100.

Additionally, following the guidelines of the Japanese Association for Infectious Diseases (JAID) and Japanese Society of Chemotherapy (JSC) for the treatment of infectious diseases [12], an antimicrobial treatment protocol was established. This involved a thorough review of the type and dosage of antimicrobial agents, standard duration of therapy, and implementation of bacteriological cultures and blood tests. This will facilitate the optimization of antimicrobial selection, dosage, and administration method during initial treatment and enable appropriate testing. At the stage of treatment decision (day 1), Plan 2 was selected if the patient was ≥65 years with serum creatinine ≤0.70 mg/dL and serum albumin <3.0 g/dL (Fig. 1) [13–15]; otherwise, the choice was made at the discretion of neurosurgeons. If the isolation rates of *P. aeruginosa* and MRSA markedly increase, the use of different types of antimicrobial agents considered annually.

2.2. Study design and patients

Patients admitted to our neurosurgery department due to stroke between January 2019 and December 2022 were included in the study. Aspiration pneumonia was diagnosed by neurosurgeons in cases of

apparent aspiration or swallowing dysfunction, accompanied by new or progressive radiographic infiltrates, and the presence of at least two of three clinical features (fever >38 °C, leukocytosis or leukopenia, and purulent discharge) [16,17]. The exclusion criteria were as follows: diagnosis of aspiration pneumonia within 48 h of admission, a history of antimicrobial use within the past 90 days, and patients receiving immunosuppressive drugs or currently undergoing immunosuppressive therapy. Patients who did not receive protocolized antimicrobial treatment from January 2019 to December 2020 were categorized as group A, and those treated according to the protocol from January 2021 to December 2022 comprised group B. The following factors were reviewed: patient age, sex, Charlson Comorbidity Index [18], underlying disease, biochemical and hematological parameters, history of admission to the intensive care unit (ICU) or stroke care unit (SCU), presence or absence of central venous catheter insertion or urinary/bladder catheter placement, severity of illness as measured by I-ROAD [19], microbiological tests performed, bacterial species detected, timing from diagnosis of aspiration pneumonia to the initiation of rehabilitation, types and dosages of antimicrobial agents, and the number of days of antimicrobial administration.

2.3. Evaluation of the efficacy of the antimicrobial treatment protocol

Patients diagnosed with aspiration pneumonia were initially treated with antimicrobial therapy and patients who changed antimicrobial agents to de-escalation or escalation as a second-line treatment due to the clinical course or microbiological tests were closely monitored until the end of their treatment. The percentage of cases in which first-choice antimicrobials were under-dosed, due to patients not receiving more than the usual dose based on renal function, and the inpatient mortality rate were compared between groups A and B. Inpatient mortality was defined as death rate occurring during the hospital admission before discharge. In addition, the cumulative number of days of antimicrobial treatment, which is the number of days from the start to the end of therapy for the same target disease, regardless of the type of antimicrobial agent, and length of hospital stay, was compared between groups A and B.

2.4. Statistical analysis

Categorical variables are reported as percentages and continuous variables are reported as medians and interquartile ranges. All statistical analyses were performed using EZR version 1.29 (Saitama Medical Center, Jichi Medical University, Saitama, Japan) [20]. Fisher's exact test (two-tailed) was used to compare categorical variables, and the

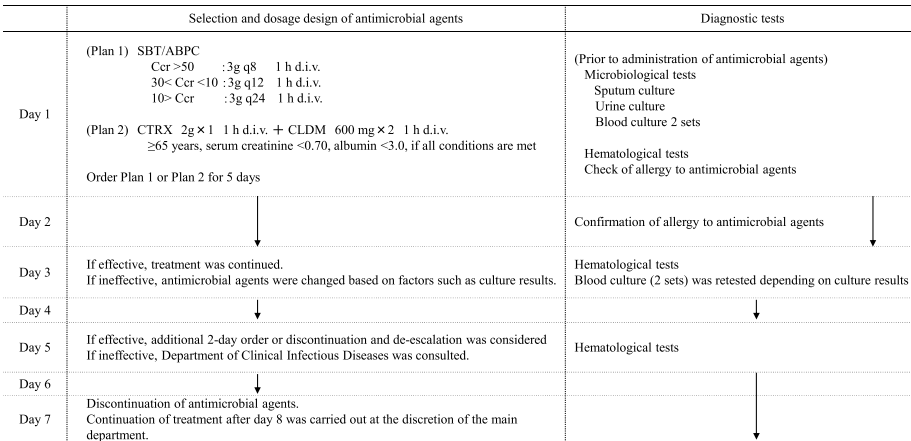


Fig. 1. Antimicrobial treatment protocol for aspiration pneumonia

Chronological order of establishment of the protocols for the selection and dosage regimen of the antimicrobial agents and microbiological and blood tests. SBT/ABPC, sulbactam/ampicillin; Ccr, creatinine clearance; d.i.v., drip infusion in vein; CTRX, ceftriaxone; CLDM, clindamycin.

Mann–Whitney *U* test was used to compare continuous variables. The log-rank test was used to compare the cumulative number of days of antimicrobial administration and the length of hospital stay. Statistical significance was set at $p < 0.050$.

3. Results

3.1. Antimicrobial treatment protocol in aspiration pneumonia

Fig. 1 illustrates the antimicrobial treatment protocol for aspiration pneumonia, developed according to the JAID/JSC Guidelines for the Treatment of Infectious Diseases, utilizing the bacterial isolation rates from respiratory specimens at our hospital. The annual isolation rates of *P. aeruginosa* from respiratory specimens from January 2018 to December 2021 were 6.5, 6.6, 7.0, and 7.0 % in JANIS and 5.5, 5.6, 6.1, and 6.0 % in our hospital. No significant changes were observed over time in MRSA isolation rates, which were 6.4 %, 6.5 %, 6.4 %, and 6.0 % nationally, and 4.4 %, 5.8 %, 7.6 %, and 7.4 % at our institution. These rates were similar to those reported for *P. aeruginosa* isolation. Based on the isolation of resistant bacteria, no change in the type of antimicrobials used in the protocol was necessary during the study period.

3.2. Patient characteristics

The study design is illustrated in Fig. 2. Group A included 66 patients and group B included 46 patients. Patient backgrounds are described in Table 1. In groups A and B, the median ages were 72.2 years (range 30–93) and 76.2 years (range 35–96), respectively. The number of males was 35 (53.0 %) in group A and 31 (67.4 %) in group B, with no significant differences between the groups. There were no significant differences in creatinine clearance, presence of dialysis, history of ICU or SCU admission, renal function, presence of central venous catheter insertion, or urinary or bladder catheter placement. However, significant differences were found in the number of sputum, urine, and blood culture sets among the microbiological tests between the two groups ($p < 0.010$). Additionally, MRSA was detected in the sputum cultures of two patients in both groups A and B. No other drug-resistant bacteria, such as extended β -lactamase-producing bacteria or multidrug-resistant *P. aeruginosa*, were detected. Additionally, no drug-resistant bacteria, including MRSA, were found in blood cultures. Tazobactam/piperacillin (TAZ/PIPC) and ceftriaxone (CTRX) were the most commonly administered antimicrobial agents in group A at the initial selection ($p < 0.050$). The percentages of second and subsequent therapies did not differ significantly between the two groups.

3.3. Efficacy of the antimicrobial treatment protocol

As shown in Fig. 3a, the percentage of patients undertreated with the first-choice antimicrobial agent was significantly lower in group B ($p < 0.050$), and the in-hospital mortality rates for groups A and B were 6.1 and 4.3 %, respectively; however, this difference was not significant (Fig. 3b). Furthermore, the cumulative number of days of antimicrobial administration in groups A and B was 9 days (95 % CI, 8–11) and 7 days (95 % CI, 6–8), respectively, while the length of hospital stay was 43 days (95 % CI, 28–55) and 28.5 days (95 % CI, 22–35), respectively, both exhibiting a significant difference ($p < 0.050$) (Fig. 3c and d).

4. Discussion

Pharmacists and physicians have collaborated to develop treatment protocols for improving the efficacy and ensuring the safety of drug therapy in various fields [21,22]. To the best of our knowledge, this is the first report on patient outcomes following protocol-based antimicrobial treatment. In an aging society such as Japan, aspiration pneumonia is expected to become a problem in the future [23,24]. In this study, we focused on older adult patients with stroke who are prone to aspiration pneumonia.

The outcome measures for antimicrobial stewardship by the AST were in-hospital mortality and length of hospital stay. Although the in-hospital mortality rates were lower in group B than in group A, the difference was not statistically significant. This lack of impact on mortality could be due to the inclusion of patients with mild to moderate aspiration pneumonia, who were not at risk for resistant organisms, and those with low underlying risks of death from aspiration pneumonia, such as dementia, peptic ulcers, and chronic lung disease [25]. We observed that adherence to the protocol led to a reduction in the median length of hospital stay from 43.0 to 28.5 days, suggesting that the absence of broad-spectrum antimicrobial administration can reduce the length of hospital stay. Although no statistically significant differences were observed, group A started rehabilitation earlier than group B. Consistent with previous reports that timely rehabilitation can lead to shorter hospital stays for patients with stroke-related pneumonia [11, 26], this factor may have contributed to the shorter hospital stays observed in this study.

In addition to considering the severity and risk factors associated with resistant organisms in patients with pneumonia [27,28], it is crucial to consider the type of causative organism and drug susceptibility in the empirical selection of antimicrobial agents. In Japan, the most important causative organisms of nosocomial pneumonia, including aspiration, are MRSA and *P. aeruginosa* [6,29,30]. The frequency of sputum-derived isolates of MRSA and *P. aeruginosa* was found

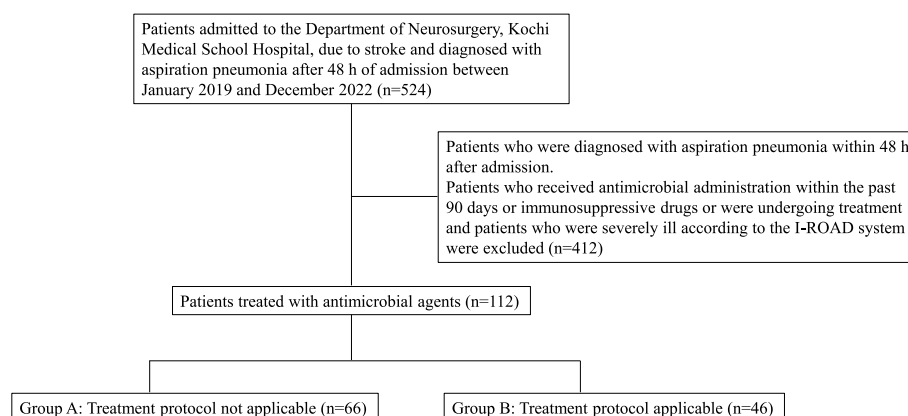


Fig. 2. Study design

Patients who did not meet the exclusion criteria were assigned to either the treatment protocol not applicable group (group A) or the treatment protocol applicable group (group B).

Table 1

Comparison of patient characteristics between groups A and B.

Variables	Group A (n = 66)	Group B (n = 46)	p-value
Age	72.2 (30–93)	76.2 (35–96)	0.082
Sex (male) (%)	35/66 (53.0)	31/46 (67.4)	0.088
Charlson Comorbidity Index (%)			
0	14/66 (21.2)	6/46 (13.1)	0.322
1–2	20/66 (30.3)	10/46 (21.7)	0.388
3–4	15/66 (22.7)	18/46 (39.1)	0.091
≥5	17/66 (25.8)	12/46 (26.1)	1.000
Severity of pneumonia by I-ROAD (%)			
Mild	32/66 (48.5)	18/46 (39.1)	0.332
Moderate	34/66 (51.5)	28/46 (60.9)	0.332
Severe	0/0 (0)	0/0 (0)	N/A
Underlying disease (%)			
Diabetes mellitus	23/66 (34.8)	18/46 (39.1)	0.693
Cardiac failure	17/66 (25.8)	16/46 (34.8)	0.400
Cerebrovascular disease	15/66 (22.7)	12/46 (26.1)	0.823
Hemiplegia (including paraplegia)	4/66 (6.1)	5/46 (10.9)	0.484
Dementia	5/66 (7.6)	3/46 (6.5)	0.571
Peptic ulcer	15/66 (22.7)	10/46 (21.7)	0.546
Solid tumor	5/66 (7.6)	8/46 (17.4)	0.138
Blood cancer	2/66 (3.0)	1/46 (2.2)	1.000
Autoimmune disorder	14/66 (21.2)	12/46 (26.1)	0.650
Chronic lung disease	16/66 (24.2)	10/46 (21.7)	0.823
Biochemical tests (range)			
Aspartate aminotransferase (U/L)	27.7 (11–79)	29.6 (9.0–164)	0.326
Alanine aminotransferase (U/L)	21.3 (5.0–141)	25.5 (6.0–141)	0.182
Serum albumin (g/dL)	3.6 (2.0–4.6)	3.3 (1.1–4.5)	0.071
C-reactive protein (mg/dL)	3.9 (0.02–44.6)	3.2 (0.01–24.0)	0.306
Creatine phosphokinase (U/L)	118 (17–378)	111 (17–453)	0.332
Blood urea nitrogen (mg/dL)	18 (6.0–60.9)	23 (5.9–129)	0.064
Serum creatinine (mg/dL)	0.82 (0.23–2.44)	0.98 (0.36–1.64)	0.548
Creatinine clearance (mL/min)	53.9 (30.1–109)	57.1 (16.7–123)	0.408
Hematological tests (range)			
Red blood cell counts (10 ⁴ /μL)	412 (249–516)	430 (250–522)	0.876
Hemoglobin (g/dL)	13.4 (4.2–16.1)	14.1 (5.9–16.3)	0.707
White blood cell counts (10 ³ /μL)	8.8 (4.1–26.2)	8.5 (3.0–21.2)	0.372
Neutrophil ratio (%)	79.1 (53–96.6)	74.8 (54–93.5)	0.079
Platelet counts (10 ⁴ /μL)	22.3 (8.8–40.1)	21.4 (6.5–40.2)	0.543
Microbiological tests (%)			
Sputum	49/66 (74.3)	46/46 (100)	<0.010
<i>Staphylococcus aureus</i>	8/49 (16.3)	10/46 (21.7)	0.340
Methicillin-resistant <i>Staphylococcus aureus</i>	3/49 (6.1)	3/46 (6.5)	0.631
<i>Staphylococcus epidermidis</i>	2/49 (4.1)	2/46 (4.3)	0.667
Coagulase-negative <i>Staphylococcus</i>	1/49 (2.0)	1/46 (2.2)	0.737
<i>Streptococcus agalactiae</i>	2/49 (4.1)	1/46 (2.2)	0.524
<i>Streptococcus salivarius</i>	1/49 (2.0)	1/46 (2.2)	0.737
<i>Streptococcus pyogenes</i>	0/49 (0)	1/46 (2.2)	0.737
<i>Streptococcus constellatus</i>	0/49 (0)	2/46 (4.3)	0.476
<i>Streptococcus pneumoniae</i>	1/49 (2.0)	2/46 (4.3)	0.476
<i>Corynebacterium</i> species	5/49 (10.2)	5/46 (10.9)	0.589
<i>Escherichia coli</i>	4/49 (8.2)	2/46 (4.3)	0.369
<i>Escherichia coli</i> ESBL	1/49 (2.0)	1/46 (2.2)	0.737
<i>Klebsiella pneumoniae</i>	9/49 (18.4)	10/46 (21.7)	0.438
<i>Klebsiella pneumoniae</i> ESBL	1/49 (2.0)	0/46 (0)	0.737
<i>Klebsiella oxytoca</i>	0/49 (0)	2/46 (4.3)	0.476
<i>Serratia marcescens</i>	1/49 (2.0)	2/46 (4.3)	0.476
<i>Pseudomonas aeruginosa</i>	2/49 (4.1)	2/46 (4.3)	0.667
<i>Acinetobacter baumannii</i>	3/49 (6.1)	2/46 (4.3)	0.530
<i>Citrobacter freundii</i>	0/49 (0)	1/46 (2.2)	0.737
<i>Enterobacter cloacae</i> complex	3/49 (6.1)	2/46 (4.3)	0.530
<i>Stenotrophomonas maltophilia</i>	2/49 (4.1)	2/46 (4.3)	0.667

Table 1 (continued)

Variables	Group A (n = 66)	Group B (n = 46)	p-value
<i>Haemophilus influenzae</i>	5/49 (10.2)	5/46 (10.9)	0.589
<i>Haemophilus parahaemolyticus</i>	3/49 (6.1)	4/46 (8.7)	0.464
<i>Candida albicans</i>	7/49 (14.3)	7/46 (15.2)	0.563
<i>Candida tropicalis</i>	0/49 (0)	1/46 (2.2)	0.737
<i>Candida glabrata</i>	1/49 (2.0)	1/46 (2.2)	0.737
<i>Candida parapsilosis</i>	0/49 (0)	3/46 (6.5)	0.285
Urine	51/66 (77.3)	46/46 (100)	<0.010
<i>Staphylococcus aureus</i>	1/51 (2.0)	1/46 (2.2)	0.737
Methicillin-resistant <i>Staphylococcus aureus</i>	0/51 (0)	1/46 (2.2)	0.726
<i>Streptococcus agalactiae</i>	0/51 (0)	1/46 (2.2)	0.726
<i>Enterococcus faecalis</i>	1/51 (2.0)	1/46 (2.2)	0.726
<i>Corynebacterium</i> species	2/51 (3.9)	2/46 (4.3)	0.651
<i>Escherichia coli</i>	6/51 (11.8)	3/46 (6.5)	0.298
<i>Escherichia coli</i> ESBL	3/51 (5.9)	2/46 (4.3)	0.549
<i>Klebsiella pneumoniae</i>	1/51 (2.0)	1/46 (2.2)	0.726
<i>Klebsiella oxytoca</i>	1/51 (2.0)	0/46 (0)	0.726
<i>Pseudomonas aeruginosa</i>	1/51 (2.0)	1/46 (2.2)	0.726
<i>Candida albicans</i>	2/51 (3.9)	2/46 (4.3)	0.651
<i>Candida tropicalis</i>	0/51 (0)	1/46 (2.2)	0.726
<i>Candida glabrata</i>	1/51 (2.0)	0/46 (0)	0.726
Blood (2 sets)	37/66 (56.1)	46/46 (100)	<0.010
<i>Staphylococcus aureus</i>	1/37 (2.7)	1/46 (2.2)	1.000
<i>Enterococcus faecalis</i>	1/37 (2.7)	0/46 (0)	1.000
<i>Enterococcus faecium</i>	1/37 (2.7)	0/46 (0)	1.000
<i>Escherichia coli</i>	1/37 (2.7)	0/46 (0)	1.000
<i>Klebsiella pneumoniae</i>	0/37 (0)	1/46 (2.2)	1.000
Central venous catheter (%)	15/66 (22.7)	8/46 (17.4)	0.329
Urinary catheter (%)	14/66 (21.2)	11/46 (23.9)	0.454
ICU or HCU admission at diagnosis of aspiration pneumonia (%)	14/66 (21.2)	6/46 (13.0)	0.196
Timing of initiation from diagnosis of aspiration pneumonia of rehabilitation (range)	4.5 (0–24)	3.8 (0–15)	0.087
Antimicrobial agents of the first-choice (%)			
SBT/ABPC	5/66 (7.6)	41/46 (89.1)	<0.010
TAZ/PIPC	24/66 (36.4)	0/46 (0)	<0.010
CEZ	8/66 (12.1)	0/46 (0)	0.080
CEZ + CLDM	1/66 (1.5)	0/46 (0)	1.000
CTRX	15/66 (22.7)	0/46 (0)	<0.010
CTRX + CLDM	0/66 (0)	5/46 (10.9)	0.042
SBT/CPZ	1/66 (1.5)	0/46 (0)	1.000
CFPM	1/66 (1.5)	0/46 (0)	1.000
MEPM	2/66 (3.0)	0/46 (0)	1.000
MEPM + VCM	3/66 (4.5)	0/46 (0)	0.643
VCM	2/66 (3.0)	0/46 (0)	1.000
LVFX	2/66 (3.0)	0/46 (0)	1.000
Antimicrobial agents of second or subsequent choice (%)	23/66 (34.8)	12/44 (27.3)	0.408

ESBL, extended-spectrum β-lactamase; ICU, intensive care unit; HCU, high care unit; SBT/ABPC, sulbactam/ampicillin; TAZ/PIPC, tazobactam/piperacillin; CEZ, cefazolin. CLDM, clindamycin; SBT/CPZ, sulbactam/cefoperazone; CFPM, cefepime. MEPM, meropenem; VCM, vancomycin; LVFX, levofloxacin.

to be low at our institution when compared with the national level, and no drug-resistant isolates were observed. These results suggested that the selection of antimicrobial agents with anti-MRSA or anti-*P. aeruginosa* activity was not necessary. Anaerobes were not detected in sputum cultures in our study because of the lack of tests aimed at their detection; however, they have been previously reported as causative agents [31,32]. Therefore, we incorporated sulbactam/ampicillin (SBT/ABPC) alone or in combination with CTRX and clindamycin in our protocol to address anaerobic activity. Given the lack of recommendations for the use of anti-anaerobic agents in the guidelines of the European Respiratory Society, European Society of Intensive Care Medicine, European Society of Clinical Microbiology and Infectious Diseases, and Latin American Thoracic Society [33] and considering previous clinical findings [34], anaerobic coverage remains open for further evaluation. The types of antimicrobial agents to be included in

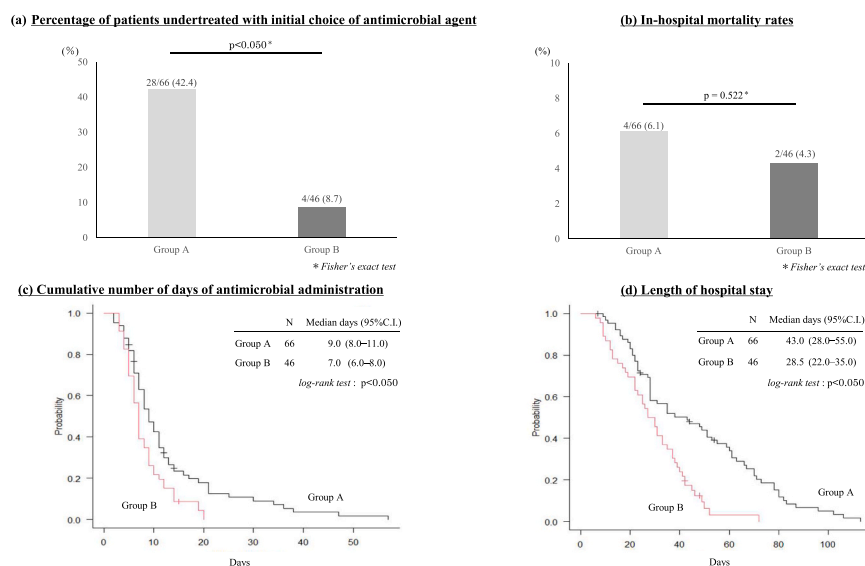


Fig. 3. Clinical outcomes with antimicrobial treatment protocol in aspiration pneumonia. Comparison of (a) percentage of patients undertreated with the initial choice of antimicrobial agent, (b) in-hospital mortality rates, (c) cumulative number of days of antimicrobial administration, and (d) length of hospital stay between groups A and B. C.I., confidence interval.

the protocol should be carefully reviewed and assessed based on previous antimicrobial susceptibility results and relevant guidelines.

In group B, all blood, sputum, and urine specimens were cultured, with only 5.4 % showing positive results for blood bacterial culture. This low culture rate is consistent with the finding of true bacteremia in approximately 7 % of community-acquired pneumonia cases [35]. Fortunately, many patients in our study did not develop bacteremia. Although performing blood cultures in patients admitted with pneumonia is considered standard care [36,37], blood culture testing may not be beneficial in patients with mild or moderate aspiration pneumonia.

In group A, TAZ/PIPC was the most common antimicrobial agent (36.4 %), and meropenem (7.5 %), a broad-spectrum antimicrobial agent, was selected as the first choice in a small number of cases. In group B, none of these antimicrobial agents were selected. Treatment with broad-spectrum antimicrobials in a population at low risk of bacterial resistance carries a potential mortality risk along with a risk of the development of resistant strains [38,39]. Sputum-derived MRSA was detected in approximately 6 % of the cases, but no other resistant bacteria were detected. Shortening the duration of antimicrobial administration in HAP can prevent the development of antimicrobial-resistance in bacteria [40]. Therefore, it is considered that the initial non-selection of broad-spectrum antimicrobial agents by the protocol and shortening the cumulative number of days of antimicrobial administration led to the prevention of drug-resistant bacteria.

The primary limitation of this study is that it is a retrospective, single-center study. Therefore, a more prospective case-control study involving multiple centers should be conducted in the future. Another limitation is the lack of a better evaluation approach for appropriate antimicrobial selection. This entails not only evaluating the efficacy and safety of the treatment but also considering the importance of protocols that enhance cost-effectiveness and reduce the workload of physicians managing infectious diseases [41].

5. Conclusions

Our results indicate that collaborative protocols agreed upon and established by physicians and pharmacists can enhance the efficacy of antimicrobial treatment for aspiration pneumonia and shorten the

duration of hospital stay, thereby improving patient quality of life. We believe that our findings will help to accelerate the involvement of hospital pharmacists in supporting the appropriate use of antimicrobial agents and serve as a framework for other facilities.

Author contributions

All the authors meet the ICMJE authorship criteria. All authors made significant contributions to various aspects of the work, including conception, study design, implementation, data acquisition, analysis, interpretation, drafting and writing of the manuscript, critical peer review, reviewing and approving changes made in the final version, and proofreading. All authors have agreed to the journal to which the paper was submitted and accept responsibility for all aspects of this research.

Ethics statement

This study was approved by the Ethical Review Board of the Kochi University School of Medicine (Registration number: ERB-107932). All the procedures adhered to the principles of the Declaration of Helsinki. Given the retrospective nature of the study, the requirement for informed consent was waived. Those who opted out of the study were excluded. The confidentiality and anonymity of patient data were maintained throughout the study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank Editage (www.editage.jp) for English language editing.

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